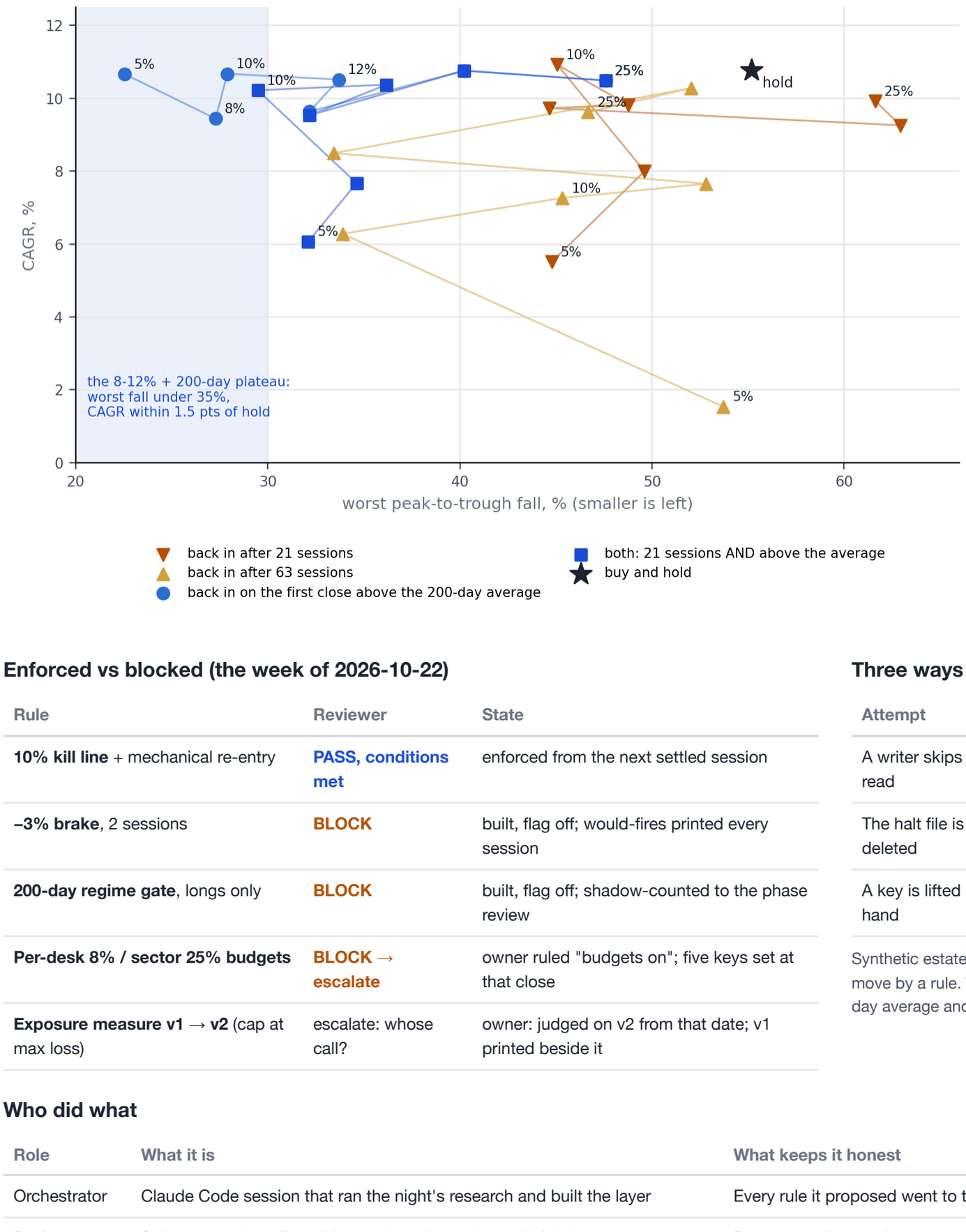
# A risk layer the agents cannot talk their way around

risk-gate · a case study in agent-governed research · Anupam Patil · 2026-10-24 · github.com/jarvisss007/risk-gate · research and engineering, not investment advice

1. **The ask.** A paper-only research estate of about eight trading desks, each run by agents, had its worst day on Thu 2026-10-22: **–4.6%** on a day the S&P 500 fell 1.2%. Its operator asked for a –3% stop, for the agents to short when the market falls, and to "make it positive".
2. **The data said no to the 3%.** On SPY 1993–2026 a 3% trailing stop turns 10.8%/yr into 3.4%/yr (re-entering after 21 sessions) and still falls 35% at worst. After a –3% day SPY is up 65% of the time over the next 21 sessions, the same as any day (65.8%). Selling the close of a bad day buys nothing.
3. **No market hedge would have helped.** SPY explains 3.4% of the account's daily P&L (beta 0.50, n=34); semiconductors 7%. Two of its three worst days were flat-SPY days. The risk was single names and option arms, and only size protects against that.
4. **The re-entry rule decides, not the threshold.** Re-entering after a fixed number of sessions walks back into 2000–02 and 2008 (10% stop: worst fall -45%). Re-entering on the first close above the 200-day average at 8–12% is one plateau: **CAGR 10.7% vs 10.8% hold, worst fall -27.9% vs -55.2%**, in both halves of the sample.
5. **A reviewer agent passed one rule and blocked three.** PASSED: the pre-registered 10% kill line with that re-entry rule. BLOCKED: a –3% brake (*"reinstates a mechanism already ruled out… he withdrew the 3% himself"*), a 200-day regime gate (*"the evidence is SPY-on-SPY… authority for the concept, not for SMA200 / month-end / longs-only"*), and per-desk budgets (*"this is his sizing ruling, not an enforcement"*). The operator made that ruling himself the next morning, "budgets on", and at that close the layer set five budget keys.
6. **The same night the reviewer found a measurement artefact.** Options counted at full delta notional with no cap at their maximum loss read net exposure as **211%** of equity; capped at max loss, **113%**. Still over the band, which is the honest finding. The fix went through the same reviewer as a versioned change to the pre-registered measure.

> **Why this is in a portfolio:** the agents did not get a prompt saying "be careful". They got a file. Every desk's writer must read one halt file before any new entry; an unreadable file refuses; a refused entry is recorded and later scored; keys set and clear by rules written down in advance; a human may hold a halt longer and never lift it earlier; nothing is ever sold to fix a number. Swap "trading desk" for "dispatch agent" or "pricing agent" and the layer is the same.

### SPY 1993–2026: a trailing stop at 5–25%, four re-entry rules




### The gate

**Settled close** · the layer computes what every rule wants; marks are not results, so it runs once, after the close

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**Reviewer agent** · read-only tools, adversarial brief · a key moves only behind a flag it passed

▼

**Halt file** · typed keys (kill, brake, regime, budget:<desk>) · every change a history row · digest over both

▼

**Writer** · reads `binding(entry)` before every new entry · a static census proves each writer has the call

▼

**Refusal** · dated record, idempotent · scored later as a forecast: did the refused entry go on to win?

▼

**Clear** · by the pre-stated rule only · a ruling may hold a key, never lift it · the high-water mark never re-arms

▼

**Owner decides** · at the gates: "budgets on" was his sizing ruling, recorded under its own id

### Enforced vs blocked (the week of 2026-10-22)

| Rule | Reviewer | State |
|---|---|---|
| **10% kill line** + mechanical re-entry | **PASS, conditions met** | enforced from the next settled session |
| **–3% brake**, 2 sessions | **BLOCK** | built, flag off; would-fires printed every session |
| **200-day regime gate**, longs only | **BLOCK** | built, flag off; shadow-counted to the phase review |
| **Per-desk 8% / sector 25% budgets** | **BLOCK → escalate** | owner ruled "budgets on"; five keys set at that close |
| **Exposure measure v1 → v2** (cap at max loss) | escalate: whose call? | owner: judged on v2 from that date; v1 printed beside it |

### Three ways around the gate, all refused ( `examples/walkthrough.sh` )

| Attempt | What happens |
|---|---|
| A writer skips the read | the census finds the function has no gate call; exit 2; the test suite fails before it ships |
| The halt file is deleted | every entry refused, shorts included: an unreadable file is never "no halt" |
| A key is lifted by hand | digest mismatch; the history row names the key that is gone with no clear; the only human move accepted is a ruling id that *holds* |

Synthetic estate, 67 sessions: 207 entries admitted, 128 refused and recorded, 3 keys set and 3 cleared, every move by a rule. The kill line tripped at –10.7%, stood 24 sessions, and cleared only when equity, the index's 200-day average and the 21-session count all agreed.

### Who did what

| Role | What it is | What keeps it honest |
|---|---|---|
| Orchestrator | Claude Code session that ran the night's research and built the layer | Every rule it proposed went to the reviewer first; three of four did not pass |
| Reviewer | Subagent with no file-editing tools and an adversarial brief | Returns verdicts and reasons, never edits; its quotes are in `docs/reviewer_verdicts.md` |
| Layer | Python, runs once per settled close | Computes what every rule wants; moves a key only behind a reviewed flag; never overwrites an unreadable file |
| Desk writers | One per desk, the only path to each book | Each reads the gate before a new entry; a static census proves it; refusals are recorded, not dropped |
| Owner | Anupam | Asked for the 3%; accepted the data's answer; made the sizing ruling himself, in his own words |

- **In the repo:** the halt file, gate, refusal record, layer, re-entry rules, census and exposure measure, written fresh and generic; the SPY stop-rule grid runner that reproduces the table from a yfinance download (28 of 29 cells to 4 decimals; the 5%/SMA cell is one fire apart, a knife edge); a synthetic estate and demo; 17 tests; the reviewer's quoted verdicts.
- **Kept out:** every account figure (equity, P&L in currency, position counts, desk names beyond generic labels, any row), the estate's code and paths, and anything that trades. The estate is paper-only; nothing here is advice or a return source.

- **Limits:** the SPY study is one index, one cost assumption (0.10%/side), one split (2010); the 10% was chosen before the grid, not fitted to it; the account's beta and $R^2$ rest on 34 sessions; the layer gates entries, never exits, so it cannot stop a loss already held.